

DAVID ROSENTHAL

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Profile Summary

Seasoned and visionary Senior Software Engineer with proven success leading the design, development, and implementation of cutting-edge software applications that drive significant revenue growth and increased business performance. Provides expertise and project oversight throughout the software development lifecycle, operating in constant compliance with regulatory guidelines, engineering design best practices, and quality standards to ensure the delivery of highly efficient software systems. A strategic problem-solver and collaborative leader, advocates for innovative and technology-advanced software solutions to elevate user experiences while achieving overarching organizational goals.

CORE COMPETENCIES

- Software Development
- Database Architecture
- UI & UX Design
- Agile Project Management
- Client & Stakeholder Engagement
- Continuous Process Improvement
- Team Leadership
- Onboarding & Mentorship
- Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Owner/Principal | Pyxis Software Design Corp.

2012 – Present

Built and manage own business providing software and UI / UX designing services for clients within the mobile and smart television spaces, including Koneksa Health, Hearst Television, Infectious Disease Group and Rise, among others. Utilize a wide range of software design and engineering tools, such as Swift, SwiftUI, Combine, Golang, Obj-C, JS / TS, React, and Redux.

Contract Lead Software Engineer | Hearst Television | New York, NY

2012 – Present

Initially hired to provide coding support, then progressed into leading numerous mobile application development projects, from design through launching. Extensive experience with GoLang for backend web services coding; Obj-C, Swift, and SwiftUI for iOS/TVOS/ WatchOS application design; MVVM design pattern with reactive programming and full Unit/UI testing; as well as with Google IMA, Google Analytics, Braze, Sentry and Conviva integrations.

- ✓ Worked alongside Project Manager, QA, and other developers to design a mobile application (iOS) for news, weather, videos, and other rich media for 28 of Hearst's TV stations. Secured approval from Apple and successfully released the software in the App Store within a year. Refactored the application to increase performance by modernizing and simplifying the code base using Swift and Combine.
- ✓ Led the development of the first iOS application for the company, one application for each of the 28 television stations in their portfolio. Worked with a team of 4 developers and successfully delivered the project on time and within budget.
- ✓ Played a crucial role in driving a substantial increase in advertising revenue due to successfully launching mobile applications. The team grew from 10 to 75 staff members.
- ✓ Developed and implemented coding standards document to streamline process, style, architecture guidelines, and rules around coding, adopted by all developers company-wide.
- ✓ Championed SwiftUI, at the early stage, influencing key stakeholders' adoption, enabling improved collaboration in shared code and more responsive design elements.
- ✓ Streamlined the build, test, and deployment phases of the application lifecycle by implementing a CI/CD pipeline, using BitRise, transitioning from manual to an automated process of unit testing, building binaries, uploading and deployment.
- ✓ Built a live streaming prototype mobile app which enabled users to stream video, in real-time, direct to a live TV broadcast. Defined key metrics to gather proof of watch time to inform pricing of advertising space, then integrated the ads into articles and videos, creating another revenue stream for the company.
- ✓ Successfully built and developed a React App for Samsung, LG, and Vizio Smart televisions. Created a containerized deployment solution and a shell script deployment solution.

Lead Software Engineer | Koneksa Health | New York, NY

2018 – 2024

Headhunted to support the development of iOS applications for collecting digital biomarkers and survey responses for pharmaceutical clinical trials. Served as the Lead Mobile engineer during the initial stages, progressively expanding accountabilities to oversee increasing iterations/changes on the application. Established a comprehensive understanding of compliance in medical software development.

- ✓ Spearheaded the design and development of the first version of the app, scheduled for a stakeholder demonstration in 10 weeks in line with an incoming clinical trial. Leveraged experience in medical-focused software development to inform the design process, secured FDA approval, and successfully launched the application on time. This resulted in increased business opportunities, and eventually the development of a patented, fully functional software integrated with a variety of medical tools for mobile phone clinical trials.

- ✓ Led other major projects, including the implementation of client-side GraphQL solution with Apollo and client-side Auth0 secure login with refresh token capability. Utilized Protobuffers, Rest APIs, Swift Realm, and numerous other 3rd party libraries using Cocoapods and SPM.
- ✓ Promoted to full-time role due to success in various projects, including providing oversight to 3 major rewrites, with the third being a vertically sliced, reactive MVVM/TD design which fast-tracked the development of a new and complete version of the software.

Software Engineer (Contract) | Infectious Disease Group | Ridgewood, NJ**2020 – 2022**

Headhunted to independently led the design, development, and integration of an iOS application enabling doctors to track patient usage of antibiotics for stewardship. The application also allowed monitoring of the patient's medical records, including location, the hospital they had been admitted, nurses in charge, diagnoses, and medicine intake/prescriptions. The project was delivered on time and under budget.

- ✓ Oversaw the design, testing, and full implementation of the iOS application, and owned all aspects of building the database infrastructure, security features, and UI and UX. Utilized an AWS EC2 instance to host a Golang REST API to an RDS instance within a MySQL database, and created an Apple enterprise distribution server, which allowed installation of iOS App outside of the App Store – a crucial feature for medical applications and privacy.

Software Engineer (Contract) | Next Level Fitness Physical Therapy | Woodcliff Lake, NJ**2019**

Built and developed an iOS application for real-time tracking of heart rate and calorie burn during exercise. Interfaced with the USDA Food Consumption database to provide nutrition data of 200k+ food items. Created the first core Bluetooth library and pioneered a framework design allowing an easy search for connecting and disconnecting heart rate monitors.

AREAS OF PROVEN PERFORMANCE

- **Software Development:** Adept at overseeing end-to-end software development lifecycle within an Agile environment, working cross-functionally with engineers, project managers, designers, UX/UI teams, and other members from design through testing and implementation, ensuring successful, on-time, and within-budget delivery of projects.
- **Project Management:** Lead a wide range of small and large-scale projects, owning all aspects of the process from requirements assessment and planning through execution, performing necessary iterations and change implementations to enhance software performance. Experience working with scrum masters through full ceremonies, scrum meetings, sprint reviews, and project planning.
- **Team Leadership:** A collaborative leader, working closely with team members to coordinate plans, monitor progress, and provide critical updates to successfully complete the project. Hired and screened potential candidates, reviewed coding exams, and onboarded and mentored junior members on best practices and project management concepts.
- **Regulatory Compliance:** Demonstrate a comprehensive understanding of software engineering best practices and industry standards, alongside extensive experience with compliance requirements for medical software development.
- **Design Tools Optimization:** Adaptable to a wide array of design and engineering tools to build applications, with specialization in iOS/iOS network-connected software and UI and UX design through Swift. Utilize Jira and Confluence for tracking and documentation, and GitHub for source code control.
- **Performance Analytics:** Establish benchmarks to measure and track target performance metrics for various applications necessary for iterations, feature enhancements, and overall software improvement.
- **Risk Identification & Mitigation:** Proven ability to assess, mitigate, and address incoming/arising issues in areas of compliance, codebase cleanliness, project management deadlines, system integration, and other key project areas.

EARLIER NOTEWORTHY EXPERIENCE**Contract Software Engineer | Rise Fitness****2012 – 2013****Contract Test System Engineer | BAE Systems****2009 – 2012****Vice President/Director, Software Engineering/Senior Software Engineer | Instrumentation Engineering****1996 – 2009****Instrumentation / Test Engineer | PRC Laser****1995 – 1996****E5 29Y/20 Shift Supervisor, Satellite Communications Ground Station Operations | US Army (Honorably Discharged)****EDUCATION & CERTIFICATIONS****Bachelor of Science, Electrical Engineering, Minor in Applied Mathematics | New Jersey Institute of Technology****TECHNICAL ACUMEN**

Swift / SwiftUI | Obj-C | JS / TS | React | Redux | Golang | iOS Applications | iPad Applications | Bitrise | GitHub Actions | MySQL Realm
HTML | REST API | GraphQL | Auth0 | Swift Protobuf | Cocoapods | SPM | JIRA | Confluence